Supplemental Materials

Metagenomic Analysis of Fecal Samples as a Differential Diagnostic Tool for Colorectal Cancer

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1 - Summary Tables

1-A: Age Summary Statistics

Table 1: Overall Age Summary

Statistic	Value
N	1643.00
Mean	61.95
Standard Deviation	11.47
Median	64.00
First Quartile (Q1)	55.00
Third Quartile (Q3)	70.00

Table 2: Age Summary by Gender

Gender	N (%)	Mean	Standard Deviation	Median	Q1	Q3
female	671 (40.8%)	62.27	11.37	64	56	70
male	972 (59.2%)	61.73	11.54	64	55	70

Table 3: Age Summary by Decade

Age (Decade)	N (%)
20-29	22 (1.3%)
30-39	46 (2.8%)
40-49	164 (10.0%)
50-59	356 (21.7%)
60-69	616 (37.5%)
70-79	391 (23.8%)
80-89	47 (2.9%)
90-99	1 (0.1%)

1-B: Age by Gender Summary Statistics

1-C: Age by Decade Summary Statistics

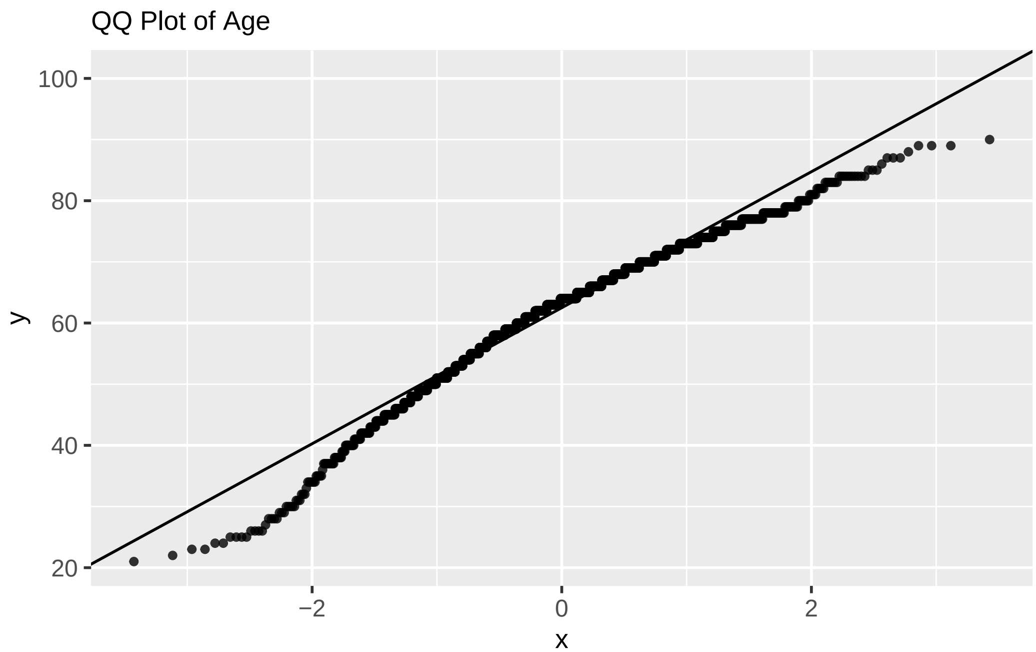
1-D: Country Counts and Proportions

Table 4: Country Overall Summary (NATO country code, trigram)

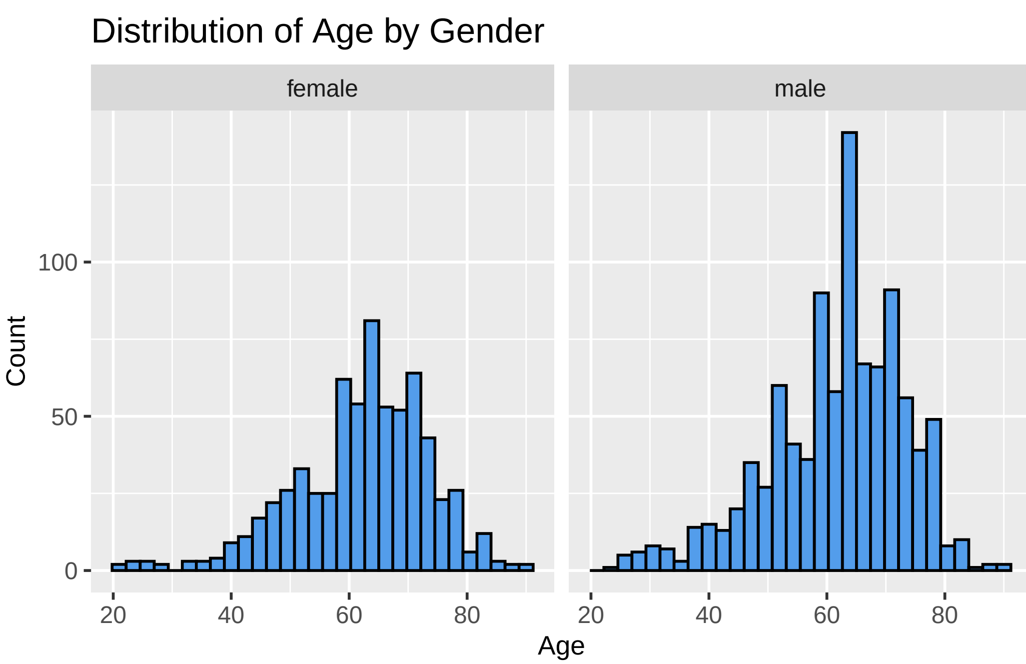
Country	N (%)
AUT	154 (9.4%)
CAN	27 (1.6%)
CHN	128 (7.8%)
DEU	125 (7.6%)
FRA	156 (9.5%)
IND	60 (3.7%)
ITA	139 (8.5%)
JPN	696 (42.4%)
USA	158 (9.6%)

2 - Supplemental Visualizations

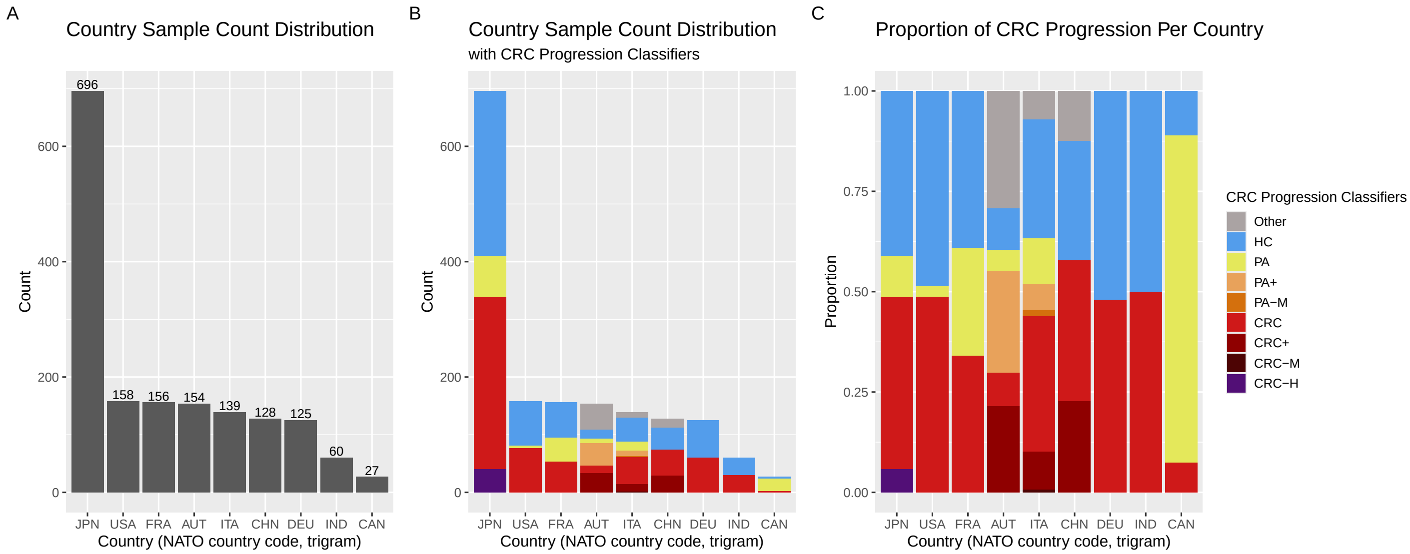
2-A: Age QQ-Plot (Supports Histogram)



2-B: Age Histogram by Gender

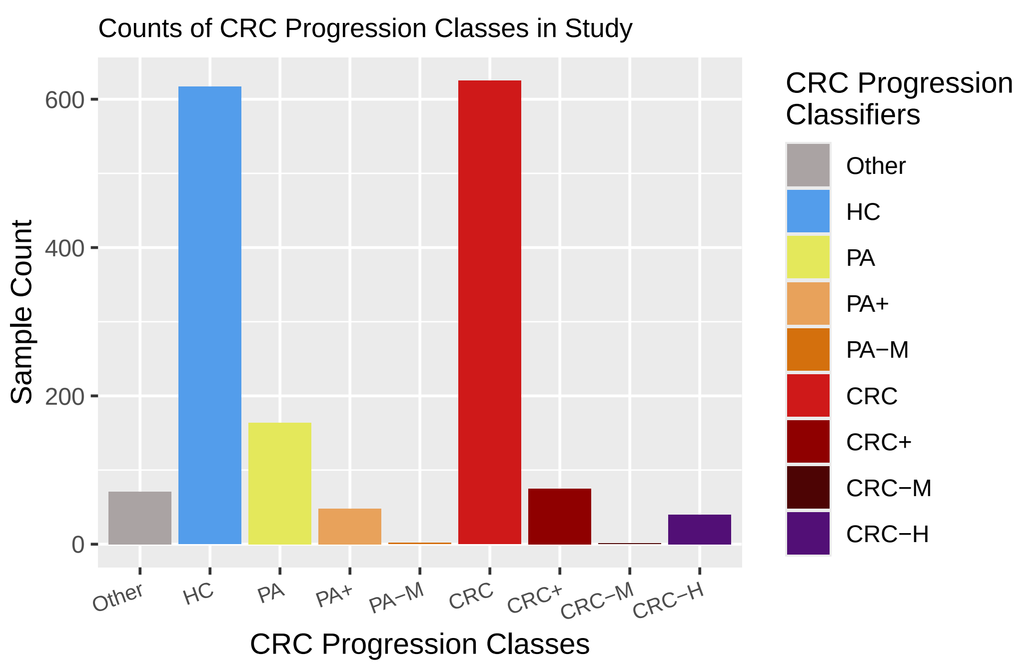


2-C: Country Distribution and Proportions



(A) Bar plot of country sample count, sorted in descending order. (B) Stacked bar plot of country sample count, sorted in descending order, filled with CRC progression classifiers. (C) Stacked bar plot of the CRC progression classifier proportion for each country in the multi−cohort study.

2-D: CRC Progression Classifier Counts



3 - Statistical Analyses

3-A: CRC Progression Classifier and Age ANOVA Significant Tukey's HSD Pairings

Disease Pairing	95% CI Lower	95% CI Upper	P Value
HC-Other	-10.50284389	-1.7236952	0.0005460
PA+-HC	1.52805911	12.0253580	0.0020679
CRC-HC	1.26185019	5.2375002	0.0000151

Disease Pairing	95% CI Lower	95% CI Upper	P Value
CRC+-HC	3.25756254	11.8241879	0.0000019
CRC+-CRC	0.01085945	8.5715406	0.0488451
CRC-H-CRC+	-14.81785403	-1.1021460	0.0097263

3-B: CRC Progression Classifier and BMI ANOVA Significant Tukey's HSD Pairings

Disease Pairing	95% CI Lower	95% CI Upper	P Value
HC-Other	-4.94428952	-1.80320222	0.0000000
PA-Other	-4.13977278	-0.57390021	0.0013977
CRC-Other	-4.53643961	-1.39667564	0.0000002
CRC-H-Other	-7.40515642	-2.45205913	0.0000000
PA+-HC	2.37307016	6.12841806	0.0000000
CRC+-HC	0.73316259	3.81667022	0.0001705
PA+-PA	1.17521532	5.29245416	0.0000411
CRC-H-PA	-4.78350442	-0.36003813	0.0095047
CRC-PA+	-5.72067642	-1.96643531	0.0000000
CRC-H-PA+	-8.48746084	-3.12375118	0.0000000
CRC+-CRC	0.32664837	3.40880795	0.0054141
CRC-H-CRC+	-6.28817020	-1.37138642	0.0000502

3-D: Linear Regression Model by CRC Progression Classifier Significant Results

Coefficients	Estimate	P Value
Other	2.793178	7.00e-07
PA	1.012194	0.004601
PA+	4.852519	4.19e-12
CRC+	2.008241	0.000225
Age centered: Other	0.158122	0.017010
Age centered: CRC-H	0.135257	0.012499

3-E: Examples of Attempted RF Models for Hyperparameter Tuning

Packages	Parameters	Classifiers	Predictors	Preprocessing	Accuracy	Confusion Matrix Results	Per -Class F1 Scores	Per-Class AUC-ROC
ranger & caret	Trees: 500	HC PA CRC Other PA+ CRC +	Taxa only	70/30 study aware manual split0.001% prevalence threshold on TRAIN, applied to all dataRemove near zero variance from TRAIN, applied to all data	Out-Of-Bag(OOB): 0.356	Accuracy: 0.4916 P-value: 9.88e-12	CRC: 0.61 HC: 0.57 All other classes showed NA	CRC 0.758 CRC+ 0.625 HC 0.709 Other 0.517 PA 0.567 PA+ 0.531

Packages	Parameters	Classifiers	Predictors	Preprocessing	Accuracy	Confusion Matrix Results	Per -Class F1 Scores	Per-Class AUC-ROC
ranger & caret	Trees: 500	HC PA CRC Other	Taxa only	70/30 study aware manual split0.001% prevalence threshold on TRAIN, applied to all dataRemove near zero variance from TRAIN, applied to all data	OOB: 0.298	Accuracy: 0.5849 P-value: 2.2e-16	CRC: 0.70 HC: 0.64 PA: 0.03 Other: NA	CRC 0.843 HC 0.757 Other 0.586 PA 0.536
ranger & caret	Trees: 500mtry: 3Target Node: 5	HC PA CRC	Taxa only	70/30 study aware manual split0.001% prevalence threshold on TRAIN, applied to all dataRemove near zero variance from TRAIN, applied to all data	OOB: 0.263	Accuracy: 0.568 P-value: 5.938e-11	CRC: 0.636 HC: 0.631 PA: NA	CRC 0.741 HC 0.689 PA 0.547
ranger & caret	Trees: 500mtry: 3Target Node: 5	HC PA CRC	Taxa only	LOSO split0.001% prevalence threshold on TRAIN, applied to all dataRemove near zero variance from TRAIN, applied to all data	Mean LOSO accuracy: 0.651 Median LOSO accuracy: 0.678	Individual plots generated rather than a data frame for averaging. This will need to be revised for future runs	Medians CRC: 0.706 HC: 0.708 PA: NA	Medians CRC: 0.809 HC: 0.763 PA: 0.562
ranger & caret	Trees: 500Target Node: 5metadata completeness threshold: 90%	HC PA CRC	Taxa, Pathway Cov, Pathway Abundance, Metadata columns that satisfy the criteria in params	70/30 study aware manual split0.001% prevalence threshold on TRAIN, applied to all dataRemove near zero variance from TRAIN, applied to all data	OOB: 0.276	Accuracy: 0.5473 P-value: 1.17e-8	CRC: 0.602 HC: 0.618 PA: NA	CRC: 0.715 HC: 0.677 PA: 0.552

4 - Relevant Links

4-A: GitHub for Script Availability

Link to our project GitHub Repository: https://github.com/chejj/Therayess

4-B: cMD3 Data Set

Link to the cMD3 data set Bioconductor Page: https://bioconductor.org/packages/3.16/data/experiment/html/curatedMetagenomicData.html